# Reweighted Atomic Norm Minimization for DOA Estimation of Mixed Coherent and Uncorrelated Signals With Coprime Arrays

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***Abstract*—In this letter, we propose a direction-of-arrival (DOA) estimation method for mixed coherent and uncorrelated signals using coprime arrays. Firstly, we generate a virtual uniform linear array (ULA) from the coprime array. Subsequently, we extract specific elements from the virtual ULA covariance matrix to obtain an augmented noise-free matrix, and reconstruct the Hermitian Toeplitz matrix by solving a reweighted atomic norm minimization (R-ANM) problem. Finally, the DOAs are estimated via MUSIC spectral search based on the reconstructed matrix. Simulation results demonstrate that the proposed algorithm outperforms existing methods.**

***Index Terms*—Direction-of-arrival (DOA) estimation, reweighted atomic norm minimization (R-ANM), coherent signals, uncorrelated signals, coprime arrays.**

## I. Introduction

DIRECTION of arrival (DOA) estimation represents a fundamental challenge in the field of array signal processing and serves as a pivotal operation for systems in radar, sonar, and wireless communications [1], [2]. Over the past several decades, various conventional algorithms have been established for scenarios involving uncorrelated signals, such as the multiple signal classification (MUSIC) algorithm [3], [4] and estimation of signal parameters via rotational invariance techniques (ESPRIT) algorithm [5]. However, the previously mentioned algorithms are ineffective for coherent signals, which typically arise in real-world environments due to multipath effects and environmental reflections. [6], [7]. Some decorrelation methods have been established to resolve this difficulty, such as the spatial smoothing (SS) approach [8], the forward-backward spatial smoothing (FBSS) technique [9] and ESPRIT-Like algorithm [10]. However, these algorithms are primarily developed for uniform linear arrays (ULAs), where the degrees of freedom (DOFs) are constrained by the count of sensors.

Coprime arrays have attracted significant attention due to their ability to generate an array with higher DOFs and a larger aperture using a limited number of sensors [11]. Zheng *et al.* in [12] developed an approach that recovers a low-rank Toeplitz matrix using nuclear norm minimization (NNM) to resolve coherent signals. In [13], a gridless algorithm is proposed for coprime arrays that recovers an augmented noise-free Hermitian Toeplitz matrix by solving an atomic norm minimization (ANM) problem, which is insensitive to the phase differences among coherent sources. The author of [14] introduced an overlapped coprime array (OCA) configuration and a low-rank matrix reconstruction model based on the Schatten-p norm to enhance consecutive lags and improve DOA estimation accuracy for coherent sources.

Furthermore, various researchers are also investigating methods to handle environments containing a mix of uncorrelated and coherent signals. In [15], the authors established a procedure centered on oblique projection, allowing independent DOA estimation for both coherent and uncorrelated sources. The row-column reconstruction of Toeplitz (RCRT) approach was proposed in [16], which builds a full-rank Toeplitz matrix using only specific rows and columns of the covariance matrix to decrease the required computational effort. A technique known as Toeplitz diagonal diffusion (TDD) was introduced in [17], which enhances estimation precision by utilizing the cross-diagonal elements of the covariance matrix. In [18], stepwise strategies are formulated for estimating the DOAs of mixed signals by utilizing the Hermitian symmetry of the covariance matrix. However, the performance of these algorithms deteriorates significantly in scenarios with closely spaced signals.

In this letter, we propose a DOA estimation method for mixed coherent and uncorrelated signals using coprime arrays. First, we generate a virtual ULA from the coprime array to enhance the DOFs. Subsequently, an augmented covariance matrix is assembled by extracting specific elements, followed by the reconstruction of a Toeplitz matrix via a reweighted atomic norm minimization (R-ANM) approach. Finally, the DOAs are resolved using the MUSIC spectrum on the recovered matrix. Simulation results demonstrate the superior performance of the proposed algorithm compared to existing techniques.

This work was supported in part by the National Natural Science Foundation of China under Grant 62501203.

*Notation*: Throughout this paper, scalars are denoted by italic letters (e.g., $x$), lowercase bold letters are reserved for vectors (e.g., $\mathbf{x}$), and uppercase bold letters are reserved for matrices (e.g., $\mathbf{X}$). The superscripts $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^H$ denote the transpose, complex conjugate, and conjugate transpose, respectively. $\text{diag}(\cdot)$ represents a diagonal matrix formed by the input vector, and $\text{tr}(\cdot)$ denotes the trace of a matrix. $\|\cdot\|_2$ and $\|\cdot\|_F$ stand for the $l_2$-norm and Frobenius norm, respectively. $\mathbb{C}^{M\times N}$ denotes the set of $M \times N$ complex matrices. $|\mathbb{P}|$ denotes the cardinality of a set $\mathbb{P}$. $\mathbf{I}_N$ represents the $N \times N$ identity matrix, and $E[\cdot]$ denotes the statistical expectation operator. $\langle\cdot\rangle_l$ denotes the $l$-th entry corresponding to the virtual position $l$.

## II. Signal Model

Consider a coprime array consisting of two sparse uniform linear subarrays. The sensor positions of the two subarrays are defined as $\{0, Md, \ldots, (N-1)Md\}$ and $\{0, Nd, \ldots, (M-1)Nd\}$, respectively, where $M$ and $N$ are coprime integers and $d = \lambda/2$ denotes the unit inter-sensor spacing, here $\lambda$ is the signal wavelength. The total number of sensors is $|\mathbb{S}| = M + N - 1$, the set of integer sensor indices $\mathbb{S}$ is given by:

$$\mathbb{S} = \{Mn\}_{n=0}^{N-1} \cup \{Nm\}_{m=0}^{M-1} \tag{1}$$

Assume that $K$ far-field narrowband signals with $\{\theta_k\}_{k=1}^{K}$ received by the coprime array. Among these $K$ signals, there are $K_c$ coherent sources and $K_u = K - K_c$ uncorrelated sources. The observed signal vector $\mathbf{x}_{\mathbb{S}}(t) \in \mathbb{C}^{|\mathbb{S}|\times 1}$ at the coprime array can be modeled as:

$$\mathbf{x}_{\mathbb{S}}(t) = \mathbf{A_c}\mathbf{s}_c(t) + \mathbf{A_u}\mathbf{s}_u(t) + \mathbf{n}_S(t) \tag{2}$$

where $\mathbf{s}_c(t) = [\mathbf{s}_1(t), \ldots, \mathbf{s}_{K_c}(t)]^T \in \mathbb{C}^{K_c}$ and $\mathbf{s}_u(t) = [\mathbf{s}_{K_c+1}(t), \ldots, \mathbf{s}_K(t)]^T \in \mathbb{C}^{K_u}$ denote the coherent and uncorrelated source vector, respectively. $\mathbf{A}_c = [\mathbf{a}(\theta_1), \ldots, \mathbf{a}(\theta_{K_c}] \in \mathbb{C}^{\mathbb{S}\times K_c}$ and $\mathbf{A}_u = [\mathbf{a}(\theta_{K_c+1}), \ldots, \mathbf{a}(\theta_K)] \in \mathbb{C}^{\mathbb{S}\times K_u}$ are the manifold matrices corresponding to coherent and uncorrelated sources, respectively. $\mathbf{a}(\theta) = \left[e^{j2\pi l_1 d\sin\theta/\lambda}, \ldots, e^{j2\pi l_{|\mathbb{S}|} d\sin\theta/\lambda}\right]^T$ is the steering vector of the $k$th source, where $l_n \in \mathbb{S}$ is the position of the $n$th sensor, and $\mathbf{n}_S(t) \sim \mathcal{CN}(0, \sigma_n^2 I)$ is the additive white Gaussian noise.

## III. Proposed method

Consider a virtual ULA denoted by $\mathbb{V} = \{0, 1, \ldots, L\}$ with $L = \max(\mathbb{S})$. Then, according to (2), the received signal vector of the virtual ULA can be expressed as:

$$\mathbf{x}_{\mathbb{V}}(t) = \mathbf{A}_{\mathbb{V}}^c\mathbf{s}_c(t) + \mathbf{A}_{\mathbb{V}}^u\mathbf{s}_u(t) + \mathbf{n}_{\mathbb{V}}(t) \tag{3}$$

where $\mathbf{A}_{\mathbb{V}}^c = [\mathbf{a}_{\mathbb{V}}(\theta_1), \ldots, \mathbf{a}_{\mathbb{V}}(\theta_{K_c})]$ and $\mathbf{A}_{\mathbb{V}}^u = [\mathbf{a}_{\mathbb{V}}(\theta_{K_c+1}), \ldots, \mathbf{a}_{\mathbb{V}}(\theta_K)]$ are the manifold matrices of the ULA corresponding to coherent and uncorrelated sources respectively. $\mathbf{a}_{\mathbb{V}}(\theta) = [1, e^{j2\pi d\sin\theta/\lambda}, \ldots, e^{j2\pi Ld\sin\theta/\lambda}]^T$ is the virtual steering vector, and $\mathbf{n}_{\mathbb{V}}(t) \sim \mathcal{CN}(0, \sigma_n^2\mathbf{I})$ is the noise vector.

The covariance matrix of $\mathbf{x}_{\mathbb{V}}(t)$ is formulated as:

$$\mathbf{R}_{\mathbb{V}} = E\{\mathbf{x}_{\mathbb{V}}(t)\mathbf{x}_{\mathbb{V}}^H(t)\} = \mathbf{A}_{\mathbb{V}}^c\mathbf{R}_{sc}\mathbf{A}_{\mathbb{V}}^{cH} + \mathbf{A}_{\mathbb{V}}^u\mathbf{R}_{su}\mathbf{A}_{\mathbb{V}}^{uH} + \sigma_n^2\mathbf{I} \tag{4}$$

where $\mathbf{R}_{sc} = \sigma_1^2\boldsymbol{\beta}\boldsymbol{\beta}^H$ is the covariance matrix of coherent signals, in which $\sigma_1^2$ is the power of the reference signal and $\boldsymbol{\beta} = [\beta_1, \ldots, \beta_{K_c}]^T$ is the complex attenuation vector, and $\mathbf{R}_{su} = \text{diag}(\sigma_{K_c+1}^2, \ldots, \sigma_K^2)$ is the covariance matrix of uncorrelated signals, and $\sigma_n^2\mathbf{I}$ denotes the noise covariance.

By right-multiplying the covariance matrix $\mathbf{R}_{\mathbb{V}}$ by a selection matrix $\mathbf{J}$, an augmented matrix $\mathbf{R} \in \mathbb{C}^{|\mathbb{V}|\times|\mathbb{S}|}$ can be obtained:

$$\begin{aligned}\mathbf{R} &= \mathbf{R}_{\mathbb{V}}\mathbf{J} \\ &= \mathbf{A}_{\mathbb{V}}^c\mathbf{R}_{sc}\mathbf{A}_c^H + \mathbf{A}_{\mathbb{V}}^u\mathbf{R}_{su}\mathbf{A}_u^H + \sigma_n^2\mathbf{J},\end{aligned} \tag{5}$$

where $\mathbf{J} = [\mathbf{e}_{l_1+1}, \mathbf{e}_{l_2+1}, \ldots, \mathbf{e}_{l_{|\mathbb{S}|}+1}]$ with $\mathbf{e}_i$ being a column vector with 1 at the $i$-th position and 0 elsewhere.

Defining the noise-free component as $\mathbf{R}_0 = \mathbf{R} - \sigma_n^2\mathbf{J}$, $\mathbf{R}_0$ can be rewritten as:

$$\mathbf{R}_0 = \mathbf{A}_{\mathbb{V}}\mathbf{B} \tag{6}$$

where $\mathbf{A}_{\mathbb{V}} = [\mathbf{A}_{\mathbb{V}}^c, \mathbf{A}_{\mathbb{V}}^u]$ and $\mathbf{B} = [(\mathbf{R}_{sc}\mathbf{A}_c^H)^T, (\mathbf{R}_{su}\mathbf{A}_u^H)^T]^T$.

According to the idea of [13], the $\mathbf{R}_0$ can be represented as a sparse sum of atoms from the set $\mathcal{A} = \{\mathbf{a}_{\mathbb{V}}(\theta)\mathbf{b}^H \mid \theta \in [-\pi/2, \pi/2], \|\mathbf{b}\|_2 = 1\}$, where $\mathbf{b} \in \mathbb{C}^{|\mathbb{S}|\times 1}$ is a unit-norm coefficient vector. The atomic $l_1$ norm $\|\mathbf{R}_0\|_{\mathcal{A}}$ provides a convex surrogate for the number of sources, which is introduced as a semidefinite programming (SDP) framework:

$$\begin{aligned}\|\mathbf{R}_0\|_{\mathcal{A}} = \inf_{\mathbf{u},\mathbf{W}} &\left\{\frac{1}{2|\mathbb{V}|}\text{tr}(\mathcal{T}(\mathbf{u})) + \frac{1}{2}\text{tr}(\mathbf{W})\right\} \\ \text{subject to} &\begin{bmatrix}\mathcal{T}(\mathbf{u}) & \mathbf{R}_0 \\ \mathbf{R}_0^H & \mathbf{W}\end{bmatrix} \succeq 0\end{aligned} \tag{7}$$

where $\mathbf{W} \in \mathbb{C}^{|\mathbb{S}|\times|\mathbb{S}|}$ is an auxiliary Hermitian matrix, $\mathcal{T}(\mathbf{u}) \in \mathbb{C}^{|\mathbb{V}|\times|\mathbb{V}|}$ is a Hermitian Toeplitz matrix with $\mathbf{u}$ being its first column.

The output of the virtual ULA is initialized as:

$$\langle\tilde{\mathbf{x}}_{\mathbb{V}}(t)\rangle_l = \begin{cases}\langle\mathbf{x}_{\mathbb{S}}(t)\rangle_i, & \text{if } l = l_i \in \mathbb{S} \\ 0, & \text{if } l \in \mathbb{V}\setminus\mathbb{S}\end{cases} \tag{8}$$

where $l \in \mathbb{V}\setminus\mathbb{S}$ represents that $l$ is an element of $\mathbb{V}$ but does not belong to $\mathbb{S}$.

The sample covariance matrix of the virtual ULA is calculated as $\hat{\mathbf{R}}_{\mathbb{V}} = \frac{1}{T}\sum_{t=1}^{T}\tilde{\mathbf{x}}_{\mathbb{V}}(t)\tilde{\mathbf{x}}_{\mathbb{V}}^H(t)$. Here, $T$ is the number of snapshots. The augmented matrix $\hat{\mathbf{R}}$ is obtained:

$$\hat{\mathbf{R}} = \hat{\mathbf{R}}_{\mathbb{V}}\mathbf{J} \tag{9}$$

In order to improve the resolution of closely spaced sources, we employ a reweighting strategy. According to [19], the optimization problem can be formulated as:

$$\begin{aligned}\min_{\mathbf{W},\mathbf{u},\mathbf{R}_0} &\frac{\mu}{2}[\text{tr}(\mathbf{Q}\mathcal{T}(\mathbf{u})) + \text{tr}(\mathbf{W})] + \frac{1}{2}\|\mathbf{R}_0 \circ \mathbf{G} - \hat{\mathbf{R}}\|_F^2 \\ \text{subject to} &\begin{bmatrix}\mathcal{T}(\mathbf{u}) & \mathbf{R}_0 \\ \mathbf{R}_0^H & \mathbf{W}\end{bmatrix} \succeq 0\end{aligned} \tag{10}$$

where $\mu$ is the regularization parameter, $\mathbf{G}$ is a matrix such that its entries are 1 at the positions corresponding to the non-zero entries of $\hat{\mathbf{R}}$, and 0 elsewhere. $\mathbf{Q} = \mathrm{diag}(q_1, \ldots, q_|\mathbb{V}|)$ is the weight matrix.

The equation (10) can be further reformulated as:

$$\min_{\mathbf{W},\mathbf{u},\mathbf{R}_0,q_i} \frac{\mu}{2}[\sum_{i=1}^{|\mathbb{V}|} q_i|\mathbf{u}_i| + \mathrm{tr}(\mathbf{W})] + \frac{1}{2}\|\mathbf{R}_0 \circ \mathbf{G} - \hat{\mathbf{R}}\|_F^2$$
$$\text{subject to} \begin{bmatrix} \mathcal{T}(\mathbf{u}) & \mathbf{R}_0 \\ \mathbf{R}_0^H & \mathbf{W} \end{bmatrix} \succeq 0 \tag{11}$$

where $\mathbf{u}_i$ denotes the $i$-th element of the first column of $\mathcal{T}(\mathbf{u})$. Clearly, the (11) is a non-convex problem due to the cross term $\sum_{i=1}^{|\mathbb{V}|} q_i|\mathbf{u}_i|$. To find the solution of (11), we propose an alternating iterative scheme, where $\mathbf{W}$, $\mathbf{u}$, $\mathbf{R}_0$, and $q_i (i = 1, \ldots, M)$ are iteratively updated. For convenience, let $P(\mathbf{W}, \mathbf{u}, \mathbf{R}_0, q_i)$ denote the objective function of (11). Then, at the $k$th iteration, we have:

$$\{\mathbf{W}^{(k)}, \mathbf{u}^{(k)}, \mathbf{R}_0^{(k)}\} = \arg\min_{\mathbf{W},\mathbf{u},\mathbf{R}_0} P(\mathbf{W}, \mathbf{u}, \mathbf{R}_0, q_i^{k-1})$$
$$\text{subject to} \begin{bmatrix} \mathcal{T}(\mathbf{u}) & \mathbf{R}_0 \\ \mathbf{R}_0^H & \mathbf{W} \end{bmatrix} \succeq 0, \tag{12}$$

and

$$q_i^{(k)} = \frac{1}{|\mathbf{u}_i^{(k-1)}| + \delta}, \tag{13}$$

where $\delta > 0$ is a small stability factor.

Based on (11), the subproblem (12) becomes a convex problem when $q_i$ is fixed, and it can be solved efficiently by CVX toolbox. After solving (11), the full-rank Toeplitz matrix $\mathcal{T}(\mathbf{u})$ can be reconstructed, and then the DOAs of mixed sources can be estimated by applying the MUSIC method.

*Remark 1:* For the convergence of the alternating iterative scheme, we have the following Theorem.

**Theorem 1.** When $q_i^{(k)} = \frac{1}{|\mathbf{u}_i^{(k-1)}|+\delta}$, the proposed alternating iterative process is convergent, guaranteeing that the objective function value decreases monotonically.

*proof.* We first introduce a log-sum function $J(\mathbf{u})$, given by:

$$J(\mathbf{u}) = \frac{\mu}{2}\sum_{i=1}^{|\mathbb{V}|} \log(|\mathbf{u}_i| + \delta) + \frac{1}{2}\|\mathbf{R}_0 \circ \mathbf{G} - \hat{\mathbf{R}}\|_F^2 + \frac{\mu}{2}\mathrm{tr}(\mathbf{W}) \tag{14}$$

Exploiting the Taylor expansion on $\log(|\mathbf{u}_i| + \delta)$, we have

$$\begin{aligned} \log(|\mathbf{u}_i| + \delta) &\leq \log(|\mathbf{u}_i^{(k-1)}| + \delta) \\ &\quad + \frac{1}{|\mathbf{u}_i^{(k-1)}| + \delta}(|\mathbf{u}_i| - |\mathbf{u}_i^{(k-1)}|) \\ &\leq q_i^{(k)}|\mathbf{u}_i| + c_i^{(k)}, \end{aligned} \tag{15}$$

where $c_i^{(k)} \geq 0$ is a constant number.

Defining a surrogate function $G(\mathbf{u})$ for $J(\mathbf{u})$ as

$$\begin{aligned} G(\mathbf{u}) &= \frac{\mu}{2}\mathrm{tr}(\mathbf{W}) + \frac{\mu}{2}\sum_{i=1}^{|\mathbb{V}|}\left(q_i^{(k)}|\mathbf{u}_i| + c_i^{(k)}\right) + \frac{1}{2}\|\mathbf{R}_0 \circ \mathbf{G} - \hat{\mathbf{R}}\|_F^2 \\ &= P(\mathbf{u}) + \frac{\mu}{2}\sum_{i=1}^{|\mathbb{V}|} c_i^{(k)} \end{aligned} \tag{16}$$

where $P(\mathbf{u})$ is the compact notation for the objective function $P(\mathbf{W}, \mathbf{u}, \mathbf{R}_0, q_i)$.

Based on the inequality (15), we obtain:

$$J(\mathbf{u}) \leq G(\mathbf{u}) \tag{17}$$

$$J(\mathbf{u}^{(k-1)}) = G(\mathbf{u}^{(k-1)}) \tag{18}$$

Since $\mathbf{u}$ is updated by solving the suboptimal problem (12), the following inequality holds:

$$P(\mathbf{u}^{(k)}) \leq P(\mathbf{u}^{(k-1)}), \tag{19}$$

which further implies

$$G_(\mathbf{u}^{(k)}) \leq G_(\mathbf{u}^{(k-1)}) \tag{20}$$

Combining (17), (18), and (20), we have:

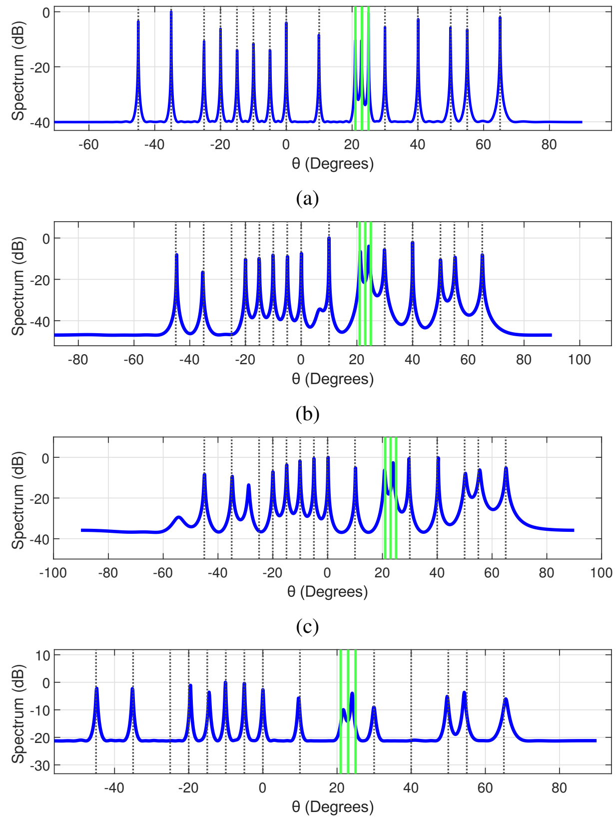
$$J(\mathbf{u}^{(k)}) \leq G(\mathbf{u}^{(k)}) \leq G(\mathbf{u}^{(k-1)}) = J(\mathbf{u}^{(k-1)}) \tag{21}$$

So, the sequence must converge to a limit point. □

## IV. Numerical Results

In this section, numerical experiments are performed to confirm the DOA estimation capabilities of the proposed algorithm within an environment consisting of mixed coherent and uncorrelated sources in the MATLAB platform. The RCRT [16], TDD [17], and 2- STEP [18] methods are also included in the experiments for comparison. For the proposed method, a coprime array with N = 7 and M = 4 is considered. Therefore, the virtual array used in the proposed method has 25 sensors, and the SDP problems are solved using CVX [20]. To ensure fairness, the setting of the array of the 2-STEP method is the same as that of the proposed method. while a ULA with 25 sensors is used for the TDD, and RCRT methods.

First, we examine the resolution performance of various algorithms in a mixed source environment. There are 17 signal sources in total, which are composed of 3 coherent signals and 14 uncorrelated signals. The three coherent sources (green lines) are positioned at 21°, 23°, and 25°, while the arrival angles for the 14 uncorrelated sources (gray lines) are established as −45°, −35°, −25°, −20°, −15°, −10°, −5°, 0°, 10°, 30°, 40°, 50°, 55°, and 65°, respectively. When the signal-to-noise ratio (SNR) is set to15 $dB$ and the number of snapshots is fixed at 500, the MUSIC spatial spectra of the compared methods are shown in Fig. 1. It can be seen that the proposed method successfully detects all the signals and other methods fail to identify all signals. This is because that the proposed method achieves superior super-resolution capability for distinguishing closely-spaced sources.




(a)

(b)

(c)

(d)

Fig. 1. MUSIC spectra of 14 uncorrelated and 3 coherent sources. (a)Proposed. (b)TDD. (c)RCRT. (d)2-STEP.

Next, we assess the precision of the analyzed algorithms for DOA estimation by calculating the Root Mean Square Error (RMSE) of the results. Additionally, the Cramér-Rao Bound (CRB) was determined to serve as a standard for performance evaluation [21]. There are 5 signal sources arriving at the array, with two coherent signals positioned at $35°$ and $30°$, while the remaining 3 uncorrelated signals are situated at $-45°$, $0°$, and $20°$. The RMSE values are determined by taking the average across 500 independent experiments.

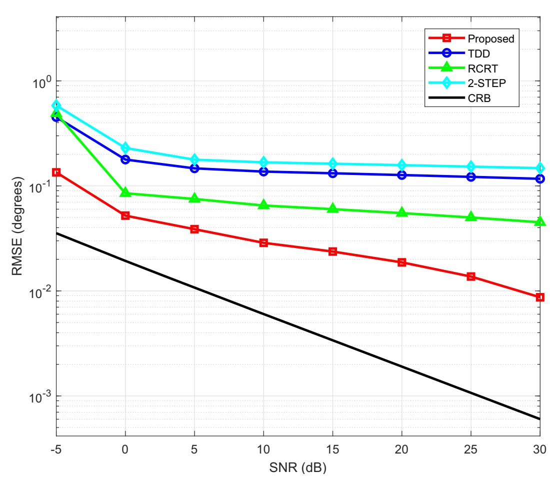



Fig. 2. RMSE comparison of various methods versus the SNR.

Fig. 2 illustrates the RMSEs for the DOA estimations of the analyzed algorithms as the SNR ranges from $-5\,dB$ to $30\,dB$ with number of samples $T = 500$. It is evident that the proposed method substantially exceeds the performance of all other methods under all SNR conditions.

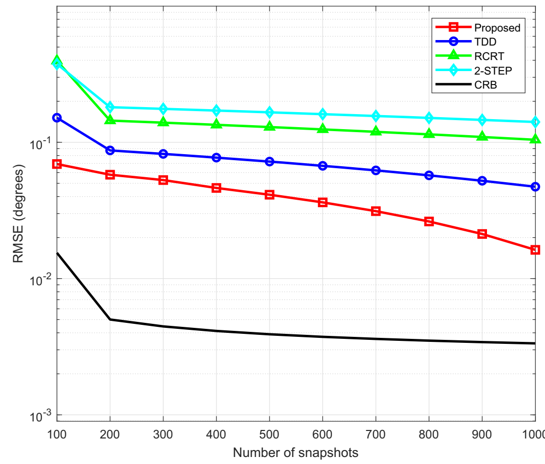



Fig. 3. RMSE comparison of various methods versus the number of snapshots.

As the count of snapshots ranges from 100 up to 1000, the RMSEs for the DOA estimations relative to the number of samples at a $10\,dB$ SNR are displayed in Fig. 3, it can be seen that the proposed method has the best performance and it is closer the CRB curve. This is mainly because the proposed method significantly reduces the inherent estimation error through reweighting.

## V. Conclusion

This letter has proposed a DOA estimation method for mixed coherent and uncorrelated signals using coprime arrays. First, a virtual ULA is generated. Subsequently, an augmented noise-free matrix is obtained by extracting specific elements from the virtual ULA covariance matrix, and the Hermitian Toeplitz matrix is reconstructed by solving a R-ANM problem. Simulation demonstrate the effectiveness and superiority of the proposed method

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